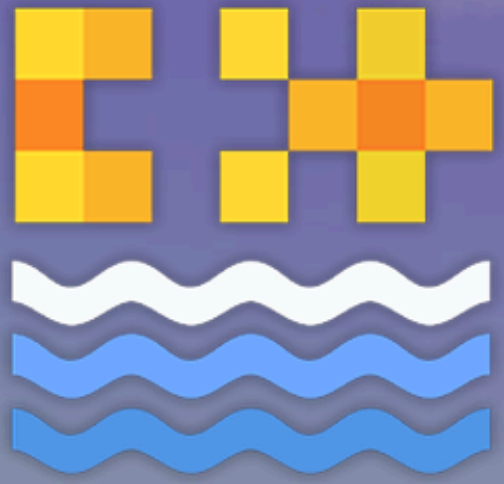




What it Takes to Implement The Standard Library II

An Excursion into libc++

Christopher Di Bella

2024

Previously on

What Does it Take to Implement the Standard Library?...

The standard library depends on compiler extensions.
Build as many C++ projects with your change as you can.
Follow software engineering best practices.
~~Don't use cppreference~~ Consult the standard directly.
Persevere.

“But *how* do I actually contribute?”

—you, probably (15 minutes ago)

Let's implement something in libc++!

```
std::ranges::fold_right
```

What is a "fold"?

```
auto const ints = std::vector<int>{43, 648, 49, 88, 110, 96, 489, 189,  
auto const expected = 1855;  
CHECK(std::accumulate(ints.begin(), ints.end(), 0) == expected);
```

What is a "fold"?

```
auto const ints = std::vector<int>{43, 648, 49, 88, 110, 96, 489, 189,  
auto const expected = 1855;
```

```
CHECK(std::accumulate(ints.begin(), ints.end(), 0, std::plus())) == exp
```

What is a "fold"?

```
auto const ints = std::vector<int>{43, 648, 49, 88, 110, 96, 489, 189,  
auto const expected = 1855;
```

```
CHECK(std::fold_left(ints.begin(), ints.end(), 0, std::plus())) == exp
```

What is a "fold"?

```
auto const ints = std::vector<int>{43, 648, 49, 88, 110, 96, 489, 189,  
auto const expected = 1855;
```

```
CHECK(std::fold_left(ints, 0, std::plus()) == expected);
```

What is a "fold"?

```
auto const ints = std::vector<int>{43, 648, 49, 88, 110, 96, 489, 189,  
auto const expected = -1855;
```

```
CHECK(stdr::fold_left(ints, 0, std::minus()) == expected);
```

Can't we just use `reverse_iterator`?

```
auto const ints = std::vector<int>{43, 648, 49, 88, 110, 96, 489, 189,  
auto const expected = -1855;
```

```
CHECK(stdr::fold_left(ints | stdv::reverse, 0, std::minus())) == expect
```

Can't we just use `reverse_iterator`?

```
auto const ints = std::vector<int>{43, 648, 49, 88, 110, 96, 489, 189,  
auto const expected = 299;  
auto right_minus = [](auto const x, auto const y) { return -y - x; };  
CHECK(stdr::fold_left(ints | stdv::reverse, 0, right_minus) == expecte
```

Relevant links

LLVM project:	https://github.com/llvm/llvm-project
libc++ documentation:	https://libcxx.llvm.org
fold_right spec:	https://eel.is/c++draft/alg.fold

Recap

Implemented `std::ranges::fold_right`

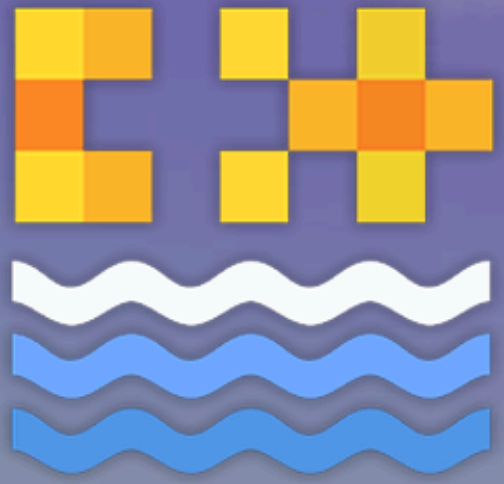
Controlling linkage visibility plays
a huge role in deep library code

Our practices *really* aren't so different to yours

_You __get __used __to __the __ugly_names

Please join us!

<https://libcxx.llvm.org/Contributing.html>



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